Best Vehicle Ever V3.0 Elegoo Car

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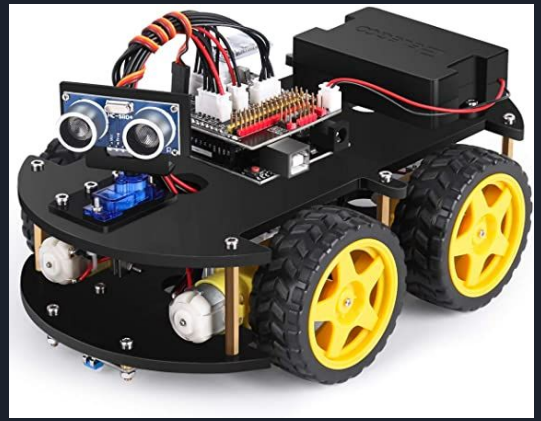
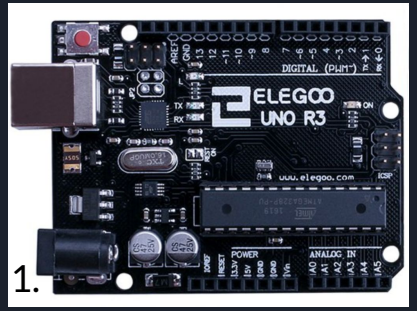


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Main Components

1. Elegoo UNO R3 Kit
2. Ultrasonic Sensor (Used for Obstacle Avoidance)
3. Servo-Motor (Used to turn the Ultrasonic Sensor)
4. Infrared Sensor (Used for Line Tracking)
5. 4x Motors
6. Battery Pack



1.



4.



2.



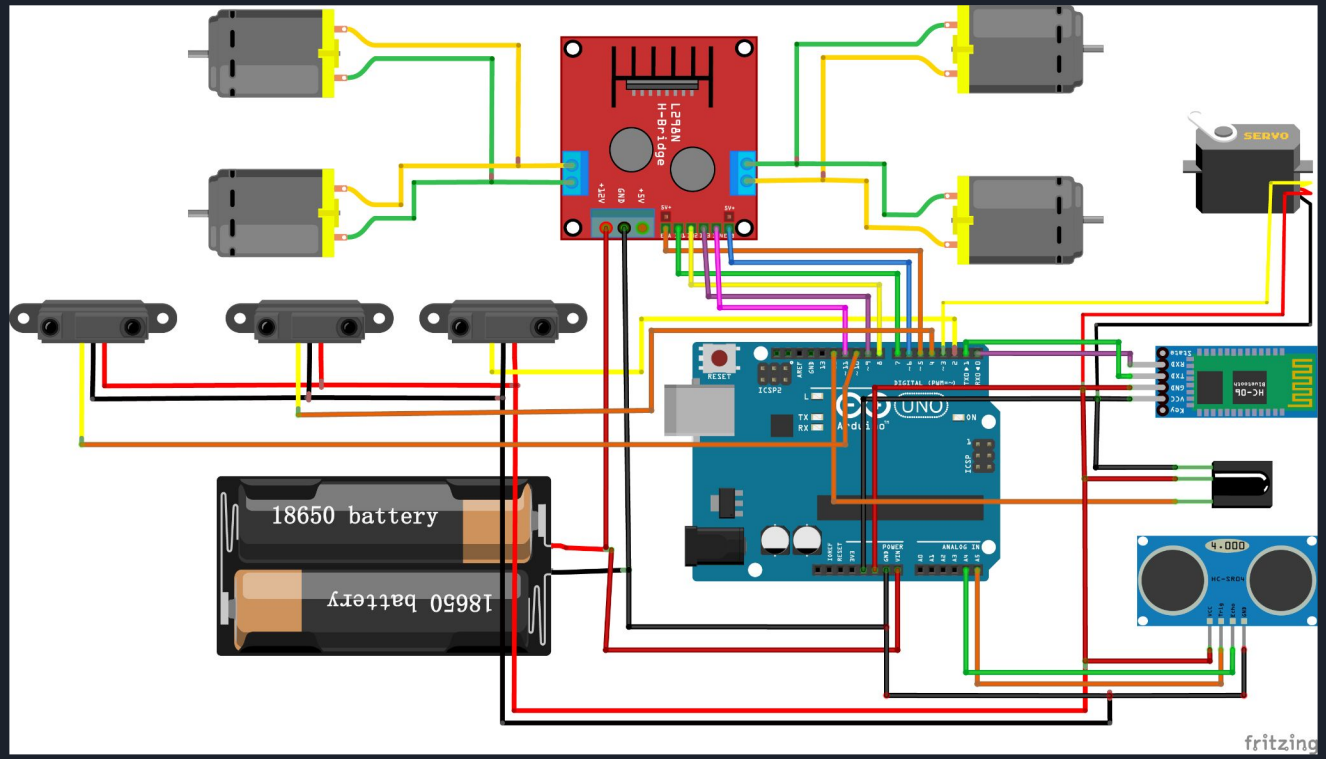
3.

How it Works

Four wheels are each connected to their respective motor that is then linked with the motor drive board (L298N). The motor drive board is then linked to the V5.0 extension shield on top of the elegoo uno R3. When powered on, the uno board will send signals to the motor drive board based on information defined in the code that was uploaded to the car. The uno R3 board is powered by a battery pack screwed to the back.



Wiring Diagram



Project Goals

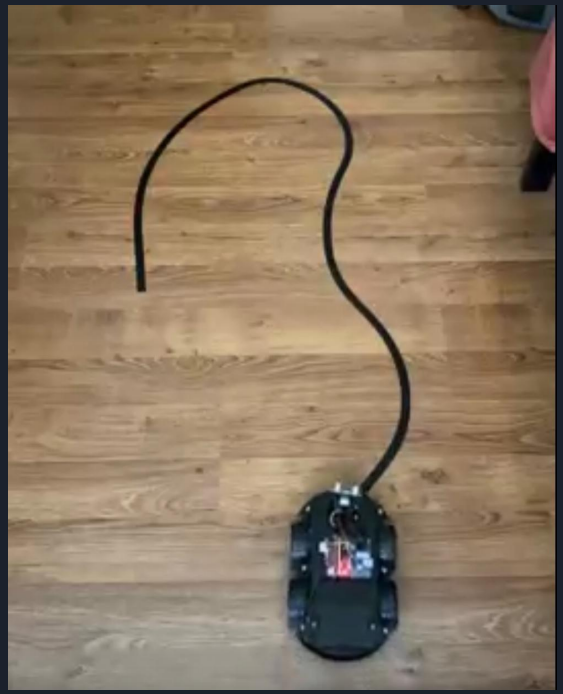
- Line Tracking with added twist
- Obstacle Avoidance

Line Tracking

Electrical Tape



Track (taken from Top View)



Problems Faced (Line Tracking):

- Uploading the code to the board itself
- There was an infinite loop when calling turn180 function
- To design a track that is compatible with the infrared sensor

Line Tracking Source Code

GitHub Link:

https://github.com/hp6164/CS113GroupProject/blob/master/GroupProject/linetracking.h

loop()

checksurround(int &c1, int &c2, int &c3)

checkValidity(int &c1, int &c2, int &c3, int TruthTable[8][3])

turn180()

Loop

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linetracking

```
}
/*********************************************************
Function Name: <loop>
 Purpose:   purpose of the function is to loop recursively again and again. Intialize the truth table, use
            a do while loop to get a valid c1, c2, c3 combinataion then performing the action according to
            c1, c2, c3 values
 Parameter: <none>
 Return: < Void function, No return value >
*********************************************************/
void loop()
{
  bool b;
  int c1 = 0;
  int c2 = 0;
  int c3 = 0;

  int TruthTable[8][3] = {
        {0,0,0}, //turn180 combination
        {0,0,1}, //Right combination
        {0,1,0}, //Left combiantion
        {0,1,1},
        {1,0,0}, //Forward combination
        {1,0,1},
        {1,1,0},
        {1,1,1},
    };

  //Loop to get a valid c1 c2 c3 combination, keep repeating the loop until the check validity function returns true
  do
  {
    checkSurround(c1,c2,c3);
    b = checkValidity(c1,c2,c3, TruthTable);

  } while (b == false);

  //Series of if and else if statements to perform the action that we associated with a few of the combinations
  //after the action is performed. loop() then gets recursively called again
  if(c1 == 1 && c2 == 0 && c3 == 0 )
  {
    forward(); //Call the forward function, to make the car go forward
    loop();
  }
  else if(c1 == 0 && c2 == 1 && c3 == 0)
  {
    left(); //Call the left function, to make the car go left
    loop();
  }
  else if(c1 == 0 && c2 == 1 && c3 == 0)
  {
    left(); //Call the left function, to make the car go left
    loop();
  }
  else if(c1 == 0 && c2 == 0 && c3 == 1)
  {
    right(); //Call the right function, to make the car go right
    loop();
  }
  else if(c1 == 0 && c2 == 0 && c3 == 0)
  {
    turn180(c1,c2,c3); //Call the turn180 function, to make the car turn 180 degrees
    loop();
  }
}
```

Done Saving.

Check Surroundings




```
/*************************************************************
Function Name: <checkSurround>
 Purpose: <purpose of the function is to check where is the line being detected>
 Parameter: int &c1, int &c2, int &c3  //These values are passed by reference, so that when c1 c2 c3 gets calculated
                                       //the values will stay changed. This is a good way to return 3 integers.
 Return: < Void function, No return value >
*************************************************************/
void checkSurround(int &c1, int &c2, int &c3)
{
  //initially c1 c2 c3 are set to 0 from the loop function

  if(LT_M) //if the line is in the middle then set c1 = 1
  {
    c1 = 1;
  }
  if(LT_L) //if the line is in the left then set c2 = 1
  {
    c2 = 1;
  }
  if(LT_R) //if the line is in the right then set c3 = 1
  {
    c3 = 1;
  }

}
```

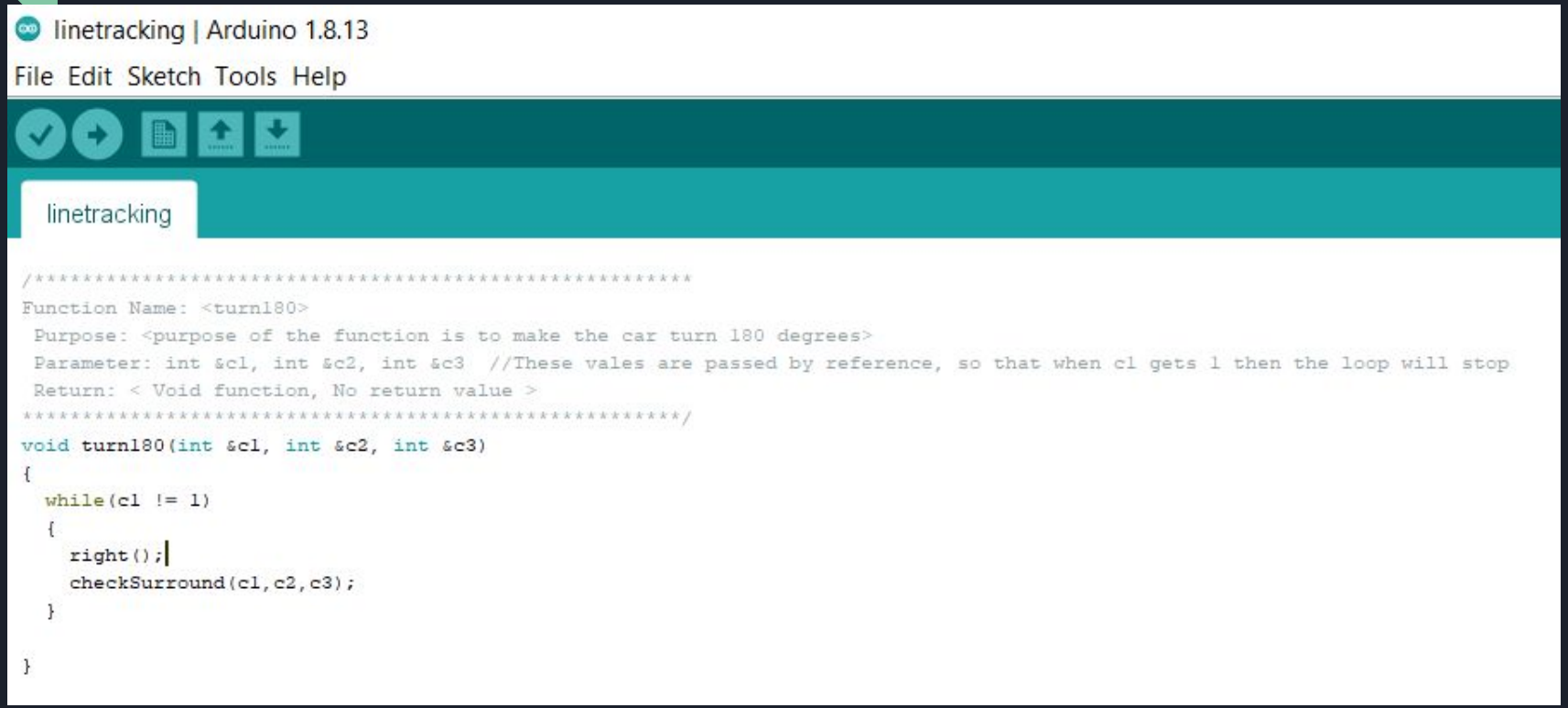
Check Validity




```
/*************************************************************
Function Name: <checkValidity>
 Purpose: <purpose of the function is to check to see if the combination of c1 c2 c3 values that gets passed into the function
           is in the truth table>
 Parameter: int c1, int c2, int c3, int TruthTable[8][3] //passed in values of c1,c2, c3 to check in the
                                                         // two dimensional array
 Return: If c1, c2, c3 matches with any combination in the truth table it will return true
         and if the combination is not in the truth table then the function will return false
*************************************************************/
bool checkValidity(int c1, int c2, int c3, int TruthTable[8][3])
{
  for(int i = 0; i < 8; i++)
  {
    if(c1 == TruthTable[i][0] && c2 == TruthTable[i][1] && c3 == TruthTable[i][2])
    {
      return true;
    }
  }

  return false;
}
```

Turn 180




```
/*************************************************************
Function Name: <turn180>
 Purpose: <purpose of the function is to make the car turn 180 degrees>
 Parameter: int &c1, int &c2, int &c3  //These vales are passed by reference, so that when c1 gets 1 then the loop will stop
 Return: < Void function, No return value >
*************************************************************/
void turn180(int &c1, int &c2, int &c3)
{
  while(c1 != 1)
  {
    right();
    checkSurround(c1,c2,c3);
  }

}
```

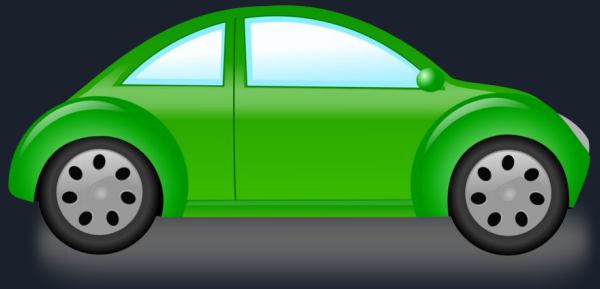
Line Tracking

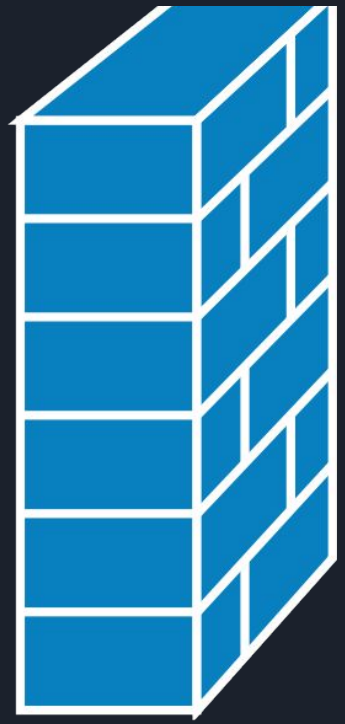


Project Goals

- Line Tracking with added twist
- Obstacle Avoidance

Obstacle Avoidance







Look Left, Look Right!!!

Problems Faced(Obstacle Avoidance):

- First attempt obstacle avoidance code didn’t work
- Car couldn’t see the obstacle at first because it was too small
- Car was turning less than 90 degrees

Source Code



GitHub Link:

https://github.com/hp6164/CS113GroupProject/blob/master/GroupProject/obsacleavoidance.h

loop()

checksurround(int &c1, int &c2, int &c3)

checkValidity(int &c1, int &c2, int &c3, int TruthTable[8][3])

Loop

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obsacleavoidance

```
/*************************************************************
* Function Name: <loop>
* Purpose: purpose of the function is to loop recursively again and again. Intialize the truth table, use
           a do while loop to get a valid c1, c2, c3 combinataion then performing the action according to
           c1, c2, c3 values
* Parameter: No parameters
* Return: < Void function, No return value >
*************************************************************/
//loop to check the surroundings c1, c2 and c3
void loop()
{
  int c1 = 0;
  int c2 = 0;
  int c3 = 0;

  int TruthTable[8][3] = {
      {0,0,0},
      {0,0,1},
      {0,1,0},
      {0,1,1},  //Forward combination
      {1,0,0},
      {1,0,1}, //Left combination
      {1,1,0}, //Right combination
      {1,1,1}, //Back combination
  };

  //Loop to get a valid c1 c2 c3 combination, keep repeating the loop until the check validity function returns true
  do
  {
    checkSurround(c1,c2,c3);
    b = checkValidity(c1,c2,c3, TruthTable);

  } while (b == false);

  //Series of if and else if statements to perform the action that we associated with a few of the combinations
  //after the action is performed. loop() then gets recursively called again
  if(c1 == 0 && c2 == 1 && c3 == 1)
  {
    forward(); //Call the forward function, to make the car go forward
    loop();
  }
  else if(c1 == 1 && c2 == 1 && c3 == 0)
  {
    right();  //Call the right function, to make the car turn right
    delay(360);
    loop();
  }
  else if(c1 == 1 && c2 == 0 && c3 == 1)
  {
    left();   //Call the left function, to make the car turn left
    delay(360);
    loop();
  }
  else if(c1 == 1 && c2 == 1 && c3 == 1)
  {
    back();    //Call the back function, to make the car go backwards
    delay(180);
    loop();
  }

}
```

Check Surroundings Pt 1

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```
/*********************************************************
* Function Name: <checksurrounding>
* Purpose: purpose of the function is to calculate c1, c2, c3 values depending upon
           what is around the car. If the middle distance is less than 30 then c1 = 1
          then the function will change the angle of the servomotor to get the distance of the
          left distance and the right distance. If the right distance is greater than left distance then
          c2 = 1, c3 = 0 other wise if left distance is greater then c2 = 0, c3 = 1. Last case is if both
          distances are less than 30 then c2=1, c3=1.
* Parameter: int &c1, int &c2, int &c3  //These values are passed by reference, so that when c1 c2 c3 gets calculated
                                        //the values will stay changed. This is a good way to return 3 integers.
* Return: < Void function, No return value >
*********************************************************/
void checkSurround(int &c1, int &c2, int &c3)
{
  int middleDistance = 0;
  int rightDistance = 0;
  int leftDistance = 0;

  myservo.write(90);  //setservo position according to scaled value
  delay(500);
  middleDistance = getDistance();

  if(middleDistance <= 30) //if middledistance is less than or equal to 30
  {
    //Then c1 gets assigned 1 and then change the angle of the servomotor to get the the distance of the right side at 10 degrees
    c1 = 1;

    stop();
    delay(500);
    myservo.write(10);
    delay(1000);
    rightDistance = getDistance();

    //Now we change the angle of the servomotor to 180 degrees to get the distance of the left
    delay(500);
    myservo.write(90);
    delay(1000);
    myservo.write(180);
    delay(1000);
    leftDistance = getDistance();

    //move the servo motor back to 90 degrees
    delay(500);
    myservo.write(90);
    delay(1000);
```

Check Surroundings Pt 2

```
    if(rightDistance > leftDistance) //if right distance is greater than left distance then c2 = 1 and c3 = 0
    {
      c2 = 1;
      c3 = 0;
    }
    else if(rightDistance < leftDistance) //else if right distance is less than c2 = 0 and c3 = 1
    {
      c2 = 0;
      c3 = 1;
    }
    else if((rightDistance <= 30) && (leftDistance <= 30)) //else if right distance or left distance is less than or equal to 30 then c2 = 1 and c3 = 1
    {
      c2 = 1;
      c3 = 1;
    }

  }
  else // if forward is greater than 30
  {
    c1 = 0;
    c2 = 1;
    c3 = 1;
  }

}
```

Check Validity

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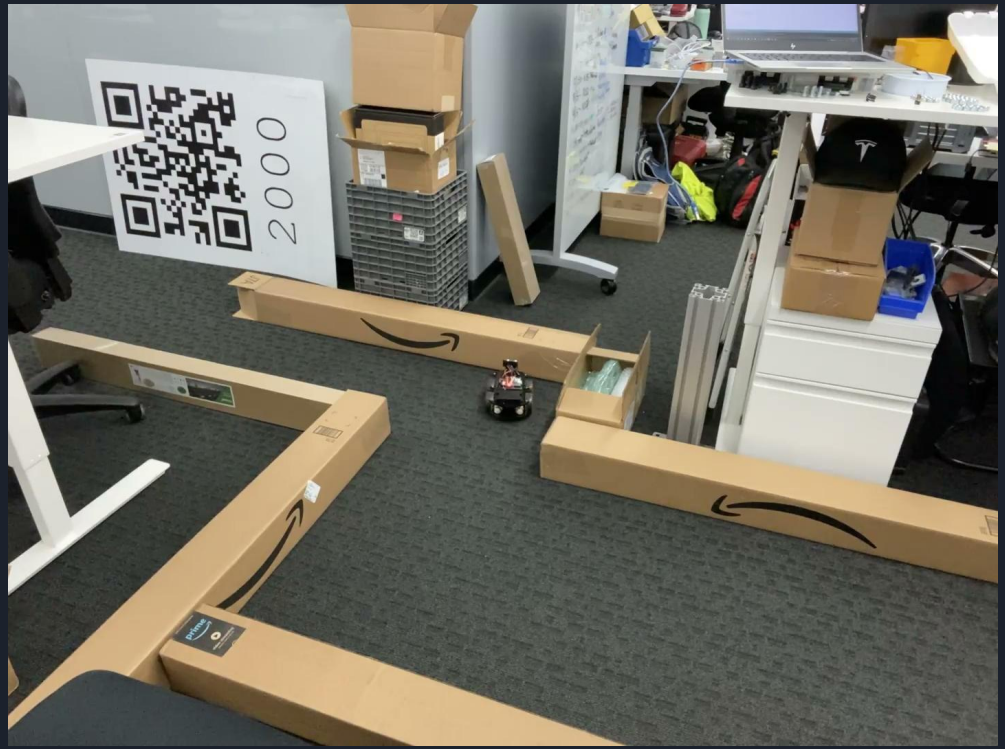
```
/*************************************************************
* Function Name: <checkValidity>
* Purpose: purpose of the function is to check the validity of the values
           c1, c2,c3 and make sure it is in the truth table

* Parameter: <c1, c2, c3 truthtable[8][3] > //passed in values of c1,c2, c3 to check in the
                                            // two dimensional array

* Return: If c1, c2, c3 matches with any combination in the truth table it will return true
          and if the combination is not in the truth table then the function will return false
*************************************************************/
bool checkValidity(int c1, int c2, int c3, int TruthTable[8][3])
{
  for(int i = 0; i < 8; i++)
  {
    if(c1 == TruthTable[i][0] && c2 == TruthTable[i][1] && c3 == TruthTable[i][2]) //if c1 c2 c3 matches at index i and each positon in the index
    {                                                                              //then return true
      return true;
    }
  }

  //if there was no match, then return false
  return false;
}
```

Obstacle Avoidance



Questions?